



B. Tech. Degree I Semester Examination December 2015

CS/EC/IT 1106 B ENVIRONMENTAL STUDIES

(2015 Scheme)

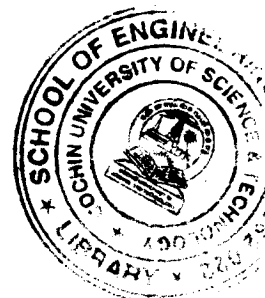
Time: 3 Hours

Maximum Marks: 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Fossil fuels are non-renewable. Why?
- (b) Mention the levels of water availability that indicate water stress and water scarcity.
- (c) Define an 'ecosystem'. What are the main components of an ecosystem?
- (d) Enumerate the factors that determine the degree of biodiversity in an ecosystem.
- (e) Suggest measures to reduce automobile emissions.
- (f) Define point and non-point sources of water pollution. Mention any two categories of water pollutants.
- (g) What is global warming and how does it occur?
- (h) What do you mean by 'sustainable development'? How can you achieve sustainable development?
- (i) How does an earthquake occur? How will you measure the severity of an earthquake?
- (j) What are the ecosystem services provided by forests?



PART B

(4 × 10 = 40)

- II. (a) How is solar energy converted to electrical power? What are the problems in this regard?
- (b) What is meant by Green Revolution? Discuss its impact on food security and environment.

OR

- III. (a) Explain the impact of deforestation on the environment.
- (b) What are the main reasons for water scarcity in India? How can we reduce the total demand for water?

- IV. (a) What is an ecological pyramid? What happens to the energy flow as we move up the trophic levels?
- (b) What are the uses of biodiversity? What are the human-induced causes of biodiversity loss?

OR

- V. (a) Distinguish between in-situ and ex-situ conservation. Explain the advantages and disadvantages of each approach.
- (b) What are tropical rainforests? Why are they important to us?

(P.T.O.)

- VI. (a) What are the major sources of noise pollution? State the effects of noise pollution on human beings.
- (b) How does ground water get polluted? What do you mean by eutrophication of lakes?

OR

- VII. (a) India is classified as a disaster prone country. Why? How does a tsunami occur?
- (b) Explain the main provisions of the following Acts.
- (i) Wildlife Protection Act of 1972.
 - (ii) Water (Prevention and Control of Pollution) Act of 1974.

- VIII. (a) Explain environmental ethics and the three different ways of viewing environmental responsibility.
- (b) What are the causes of ozone layer depletion? Mention the objectives of Montreal protocol.

OR

- IX. (a) What do you mean by the term 'population explosion'? How is population growth measured?
- (b) Discuss the problems faced by a child working as a bonded labourer in a hazardous industry.